

Ex 6 : Encoder and Decoder

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CSE

L2T1

Answer to the Question No. 3

$$f = \Sigma (1, 2, 4, 6, 8, 12)$$

Truth table:

w	x	y	z	f
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Equation:

$$f = m_1 + m_2 + m_4 + m_6 + m_8 + m_{12}$$

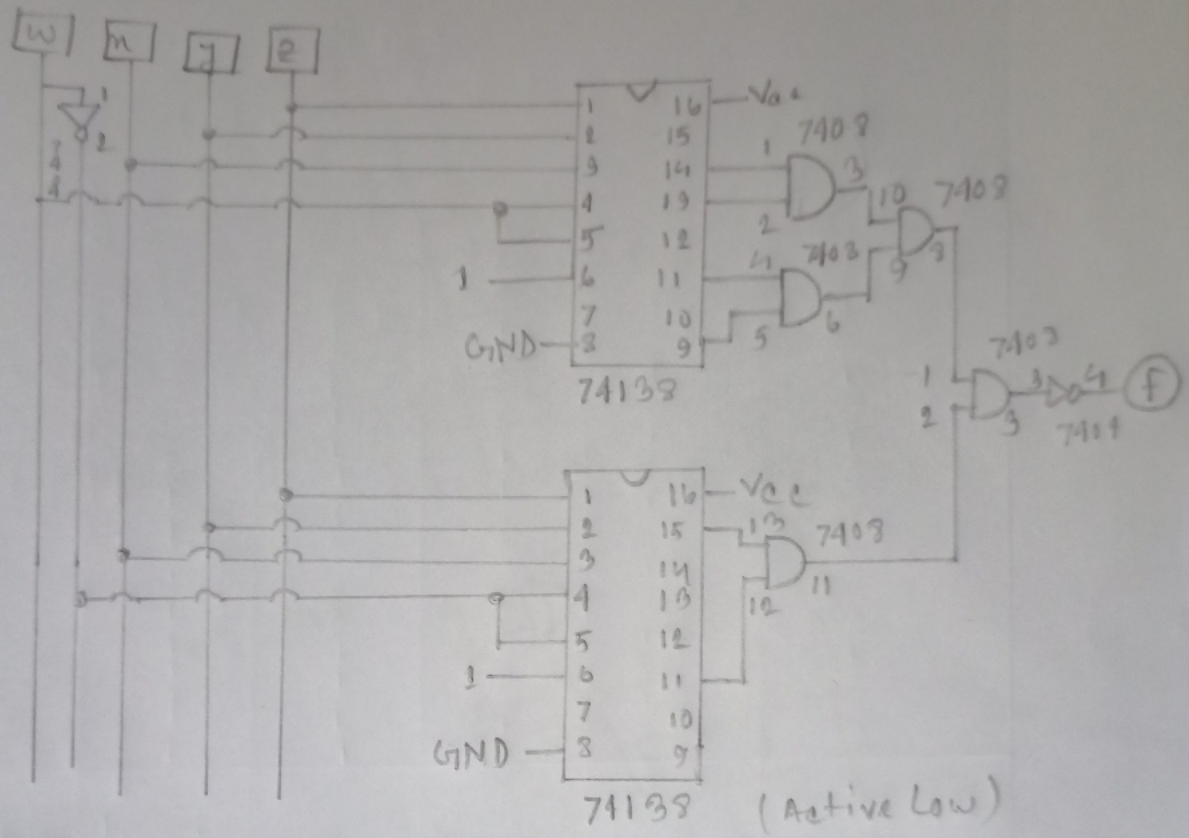
$$= \overline{m_1 + m_2 + m_4 + m_6 + m_8 + m_{12}}$$

$$= \overline{m_1 + m_2 + m_4 + m_6}$$

$$= \overline{m_1 m_2 m_4 m_6 m_8 m_{12}}$$

$$= M_1 M_2 M_4 M_6 M_8 M_{12}$$

Circuit Diagram:



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